Low Power Design

A1) Power Dissipation in CMOS
A2) Design Techniques
A3) Trends

Power Dissipation in CMOS

comes from two sources:

1. **Dynamic power dissipation** due to
   - charging and discharging of load capacitances

2. **Static power dissipation** due to
   - subthreshold current through OFF transistors
   - Tunneling current through the gate oxide

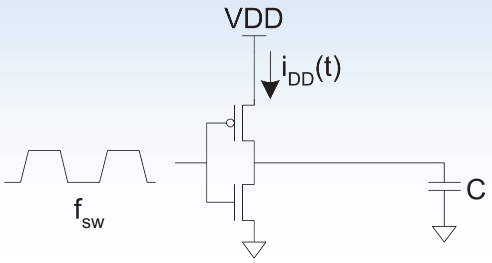
Power Dissipation in CMOS

$$P_{total} = P_{dynamic} + P_{static}$$

Dynamic Power

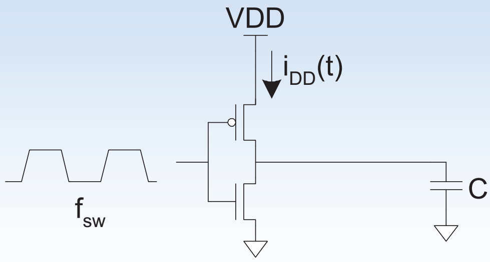
- Dynamic power is required to charge and discharge load capacitances when transistors switch.

- One cycle involves a rising and falling output.
- On rising output, charge $Q = CV_{DD}$ is required
- On falling output, charge is dumped to GND
- Power depends on switching frequency f_{sw} of input



Dynamic Power Cont.

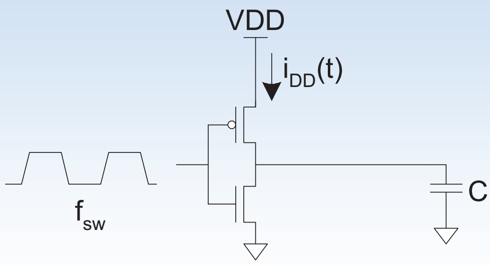
$$P_{\text{dynamic}} =$$



Low Power Design

Dynamic Power Cont.

$$\begin{aligned} P_{\text{dynamic}} &= \frac{1}{T}\int_0^T i_{DD}(t)V_{DD}dt \\ &= \frac{V_{DD}}{T}\int_0^T i_{DD}(t)dt \\ &= \frac{V_{DD}}{T}\left[Tf_{\text{sw}}CV_{DD}\right] \\ &= CV_{DD}{}^2 f_{\text{sw}} \end{aligned}$$



Activity Factor

- Suppose the system clock frequency = f
- Let $f_{sw} = \alpha f$, where α = activity factor
  – If the signal is a clock, $\alpha = 1$
  – If the signal switches once per cycle, $\alpha = ½$
  $\alpha = ½$ is max. for static CMOS logic
  – Static CMOS logic:
    - Depends on design, but typically $\alpha = 0.1$
- Dynamic power: $P_{\text{dynamic}} = \alpha C V_{DD}^{2} f$

Example

- 200 M transistor chip
  – 20M logic transistors
    - Average width: 12 λ, $\lambda * 2 =$minimum feature size
  – 180M memory transistors
    - Average width: 4 λ
  – 1.2 V 100 nm CMOS process: $\lambda * 2 =$100 nm
  – C_g = 2 fF/μm

Dynamic Example

- Static CMOS logic gates: activity factor = 0.1
- Memory arrays: activity factor = 0.05 (many banks!)
- Clock frequency f=1 GHz

- Estimate dynamic power consumption! Neglect wire capacitance and short-circuit current.

Dynamic Example

- Static CMOS logic gates: activity factor = 0.1
- Memory arrays: activity factor = 0.05 (many banks!)
- Clock frequency f=1 GHz

- Estimate dynamic power consumption!

$$C_{\text{logic}} = \left(20\times10^{6}\right)\left(12\lambda\right)\left(0.05\mu m/\lambda\right)\left(2\,fF/\mu m\right) = 24nF$$

$$C_{\text{mem}} = \left(180\times10^{6}\right)\left(4\lambda\right)\left(0.05\mu m/\lambda\right)\left(2\,fF/\mu m\right) = 72nF$$

$$P_{\text{dynamic}} = \left[0.1C_{\text{logic}} + 0.05C_{\text{mem}}\right]\left(1.2\right)^{2} f = 8.6 \text{ mW/MHz}$$

$$P_{dynamic} = 8.6\; mW/MHz \cdot 1\; GHz = 8.6\; W$$

Static Power

due to

- subthreshold current through OFF transistors

$$I_{subthreshold\_leakage} = I_{ds0} \cdot e^{\frac{V_{gs}-V_t}{n \cdot v_T}} \cdot (1 - e^{\frac{-V_{ds}}{v_T}})$$

subthreshold current increases with reduced threshold voltage of MOS transistor!

typ. 20 nA/µm for low V_t MOS trans.
0.02 nA/µm for high V_t MOS trans.

Static Power

due to

- Tunneling current through the gate oxide

$$I_{gate\_leakage} = f(gate\_oxide\_thickness)$$

typ. 3 nA/µm for thin oxide
0.002 nA/µm for thick oxide

$$I_{static} = I_{subthreshold\_leakage} + I_{gate\_leakage}$$

Leakage Example

- 200 M transistor chip
  - 20M logic transistors
    - Average width: 12 λ, $\lambda * 2 =$ minimum feature size
  - 180M memory transistors
    - Average width: 4 λ
  - 1.2 V 100 nm CMOS process: $\lambda * 2 =$ 100 nm
  - Process offers two transistor types:
    high threshold/thick oxide thickness: slow
    low threshold/thin oxide thickness: fast

Leakage Example

❑Subthreshold leakage:

–20 nA/μm for low V_t

–0.02 nA/μm for high V_t

❑Gate leakage:

–3 nA/μm for thin oxide

–0.002 nA/μm for thick oxide

❑Memories use low-leakage transistors everywhere

❑Gates use low-leakage transistors on 80% of logic

Leakage Example Cont.

- Estimate static power:

Leakage Example Cont.

- Estimate static power:
  - High leakage: $\left(20\times10^{6}\right)(0.2)(12\lambda)(0.05\mu m/\lambda)=2.4\times10^{6}\,\mu m$
  - Low leakage: $\left(20\times10^{6}\right)(0.8)(12\lambda)(0.05\mu m/\lambda)+$ $\left(180\times10^{6}\right)(4\lambda)(0.05\mu m/\lambda)=45.6\times10^{6}\,\mu m$

$$
\begin{aligned}
I_{static} &= \left(2.4\times10^{6}\,\mu m\right)\left[(20nA/\mu m)/2+(3nA/\mu m)\right]+\\
&\quad \left(45.6\times10^{6}\,\mu m\right)\left[(0.02nA/\mu m)/2+(0.002nA/\mu m)\right]\\
&= 32mA\\
P_{static} &= I_{static}V_{DD}=38mW
\end{aligned}
$$

Leakage Example Cont.

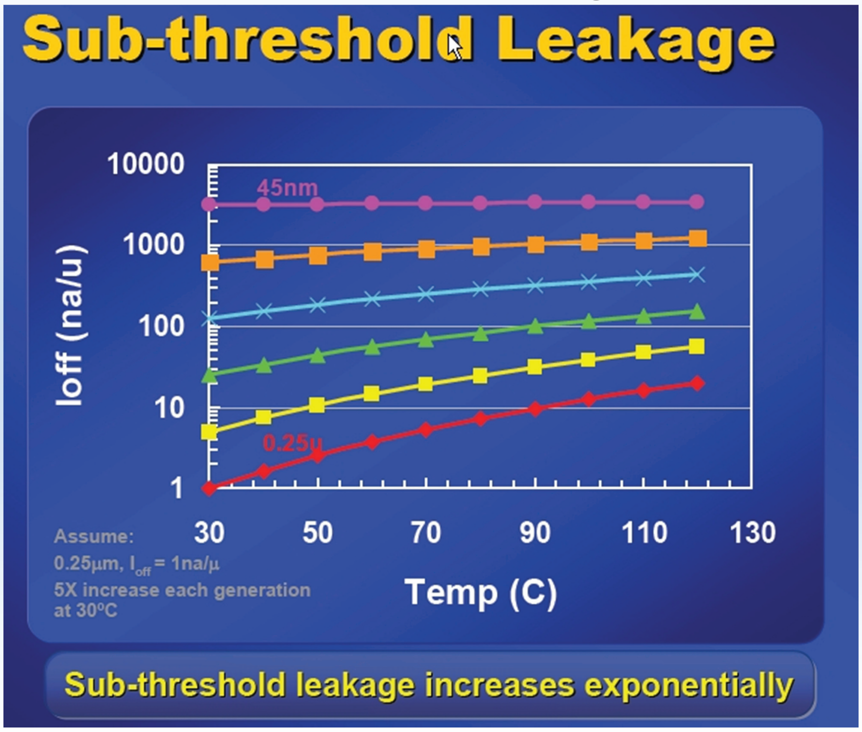
- Estimate static power:
  - High leakage: $(20\times10^6)(0.2)(12\lambda)(0.05\mu m/\lambda) = 2.4\times10^6\,\mu m$
  - Low leakage: $(20\times10^6)(0.8)(12\lambda)(0.05\mu m/\lambda) +$ $(180\times10^6)(4\lambda)(0.05\mu m/\lambda) = 45.6\times10^6\,\mu m$

$$\begin{aligned} I_{static} &= (2.4\times10^6\,\mu m)\left[(20nA/\mu m)/2 + (3nA/\mu m)\right] + \\ &\quad (45.6\times10^6\,\mu m)\left[(0.02nA/\mu m)/2 + (0.002nA/\mu m)\right] \\ &= 32mA \end{aligned}$$

$$P_{static} = I_{static}V_{DD} = 38mW$$

- If no low leakage devices used: P_{static} = 749 mW (!)

Sub-threshold Leakage Problem

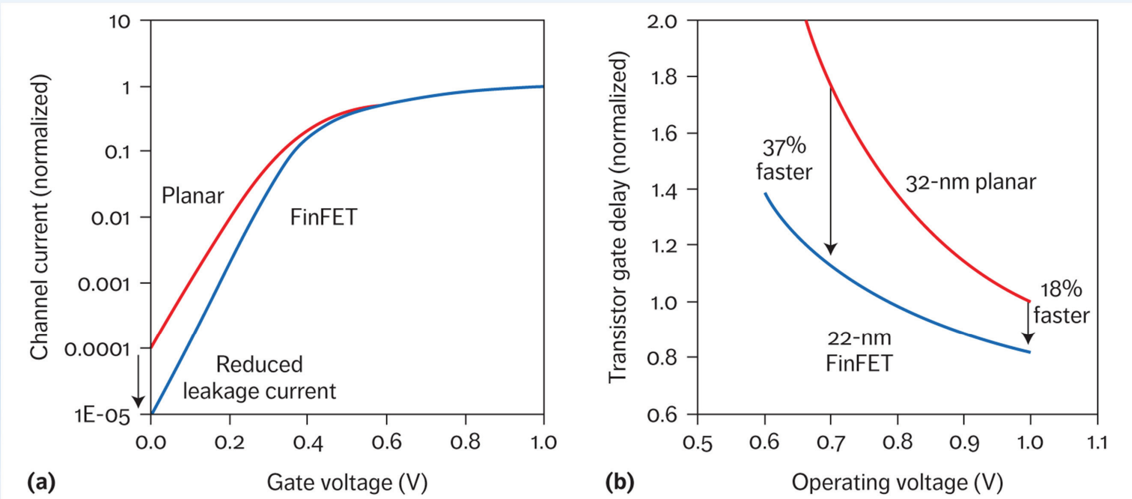


[Intel Labs 04]

Challenge: Low Power

- ❑V_{DD} decreases
  - –Save dynamic power
  - –Protect thin gate oxide
- ❑V_t must decrease to maintain device performance
- ❑But this causes exponential increase in OFF leakage
- ❑Static power dissipation is the major future challenge

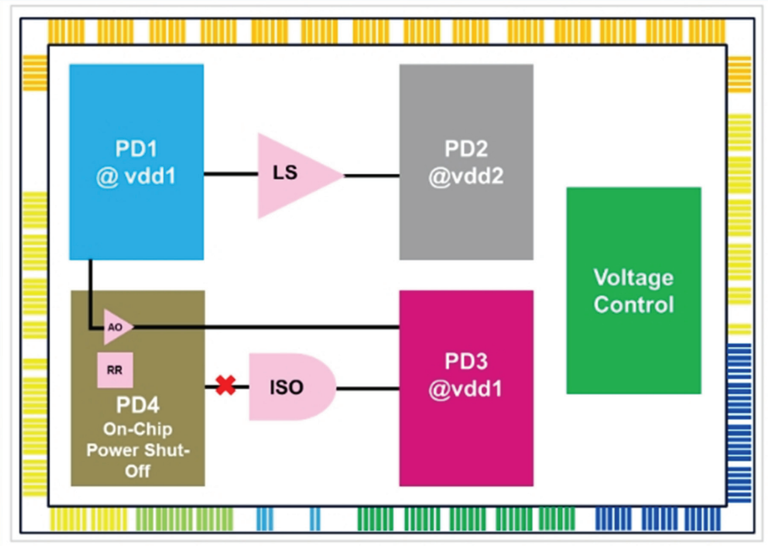
Beyond planar CMOS



[IEEE 2017]

FinFET got introduced with 22nm technology

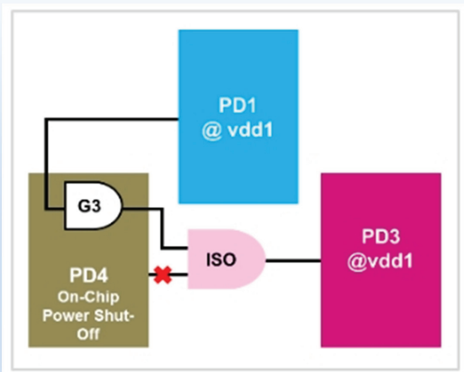
Multi VDD



[ST Microelectronics 2015]

Different functional blocks run at a different supply voltage

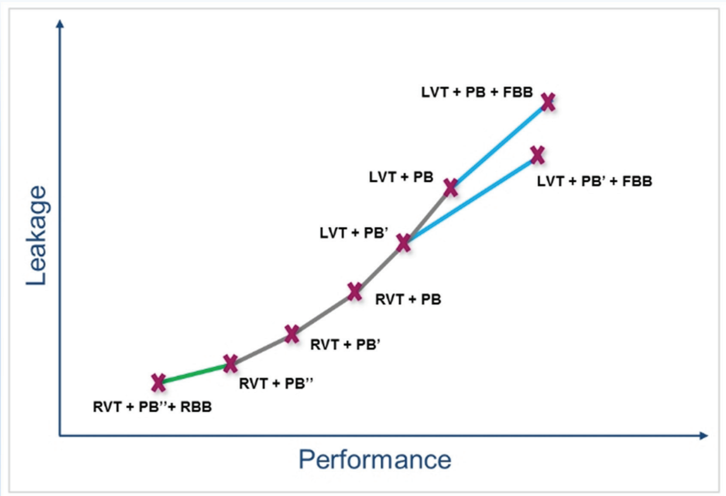
Power Gating



[ST Microelectronics 2015]

Leakage power of the CMOS circuit can be reduced by shutting down the block/module for a particular interval of time. Isolation cells (ISO) are required in the interface between shut-down and powered-up blocks to ensure that there are no floating inputs to active powered up blocks.

Multi-Threshold Voltage Support



[ST Microelectronics 2015]

To strike the best trade-off between leakage power and speed, Multi-Threshold-Voltage libraries are offered with Regular-VT(RVT)and Low-VT(LVT)

Trends

- Self-timed Design
  Idea: eliminate the clock entirely

- Wave Pipelining
  Idea: eliminate the sequencing elements

Self-timed Design

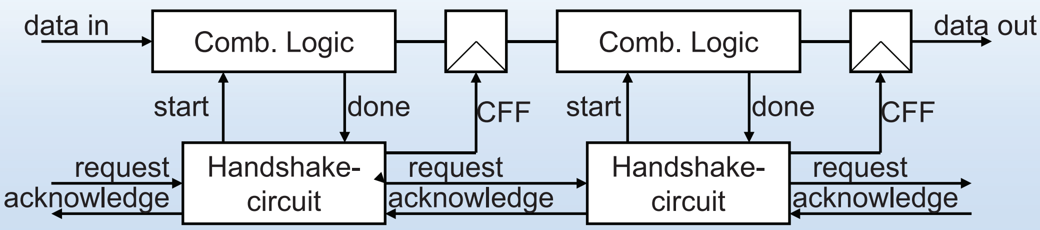
Functions of clock in synchronous design

1) Acts as completion signal
2) Ensures the correct ordering of events

Self-timed design

1) Completion ensured by completion signal
2) Ordering imposed by handshaking protocol

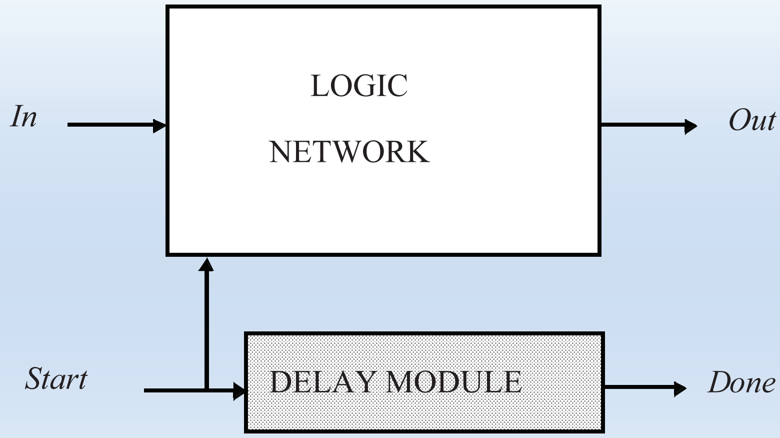
Self-timed Pipelined Datapath



Handshake-circuit: implementation of the handshaking protocol

Request:
indicates that new data is valid to be processed by comb. logic
Acknowledge:
when computation is done handshake-circuit sends a control signal CFF to flip-flop and an acknowledge to previous pipeline

Completion Signal Generation

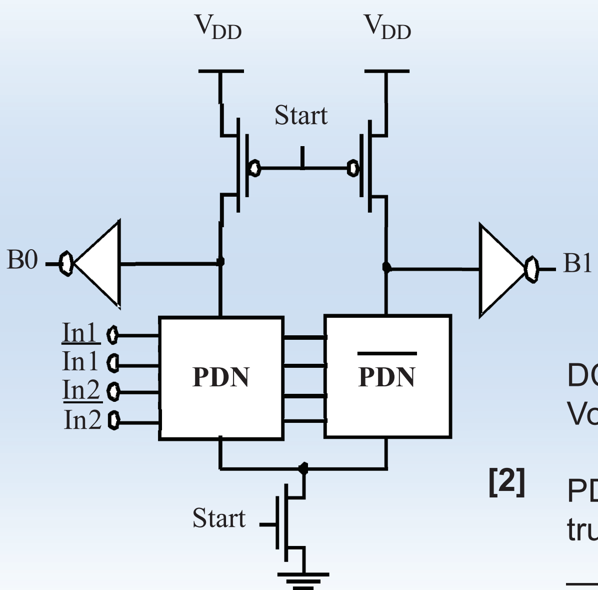


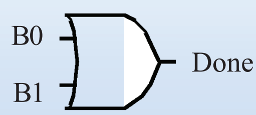
[2]

Using Delay Element (e.g. in memories)

Delay Module = max. delay of Logic Network

Completion Signal in DCVSL





[2]

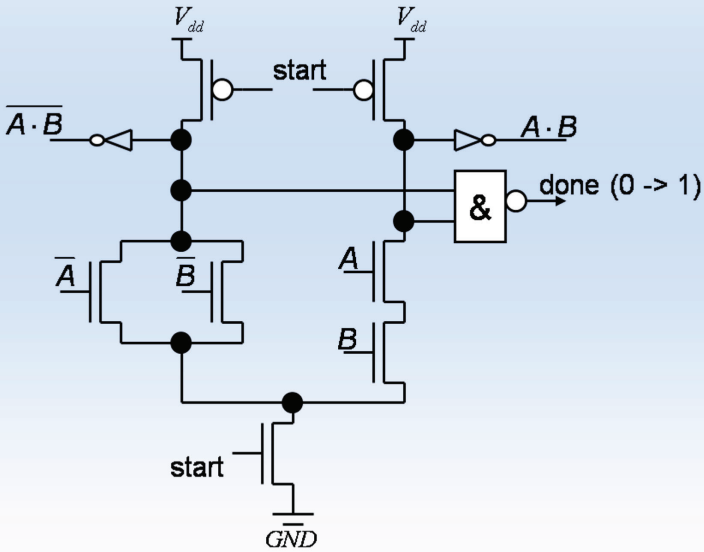
DCVSL: Dynamic Cascode Voltage Switch Logic

PDN: Pull Down Network with true inputs

$\overline{\text{PDN}}$: Complementary Pull Down Network with invers inputs

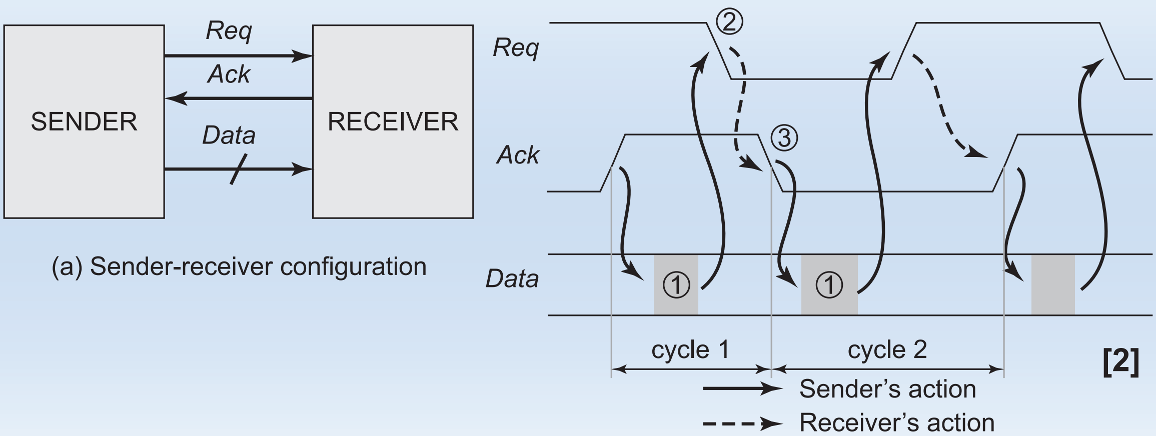
Completion Signal in DCVSL

Ex: NAND-Gate



Handshaking Protocol

Two Phase Handshake

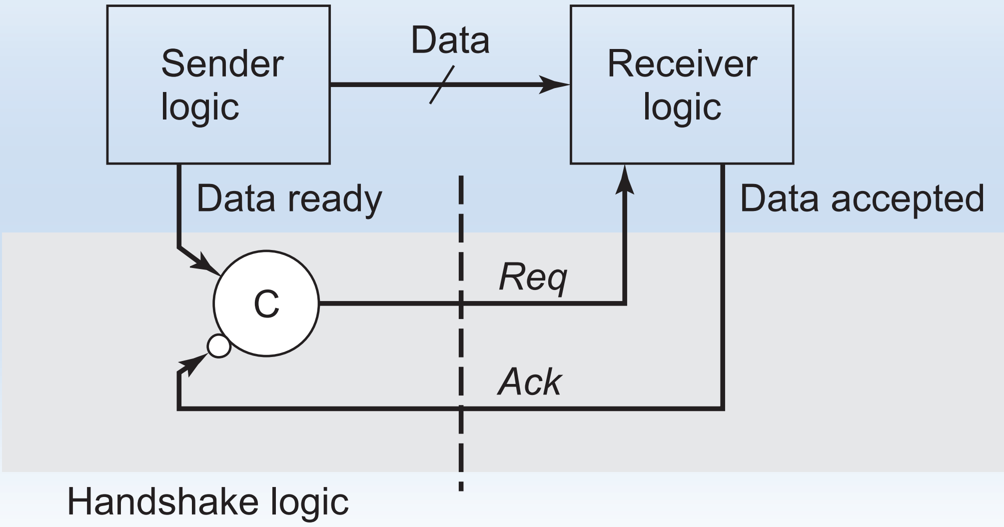


(a) Sender-receiver configuration

(b) Timing diagram

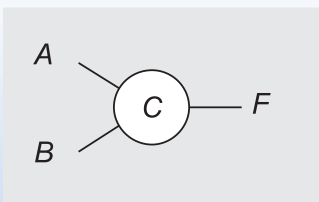
Handshaking Protocol

Two Phase Handshake



[2]

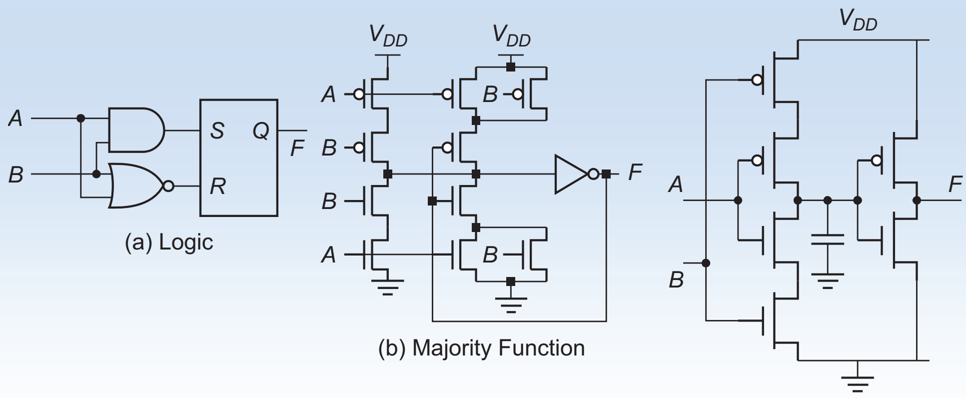
Handshake Logic: Muller-C Element



(a) Schematic

A	B	F_{n+1}
0	0	0
0	1	F_n
1	0	F_n
1	1	1

(b) Truth table



(a) Logic

(b) Majority Function

(c) Dynamic

[2]

Asynchronous Companies

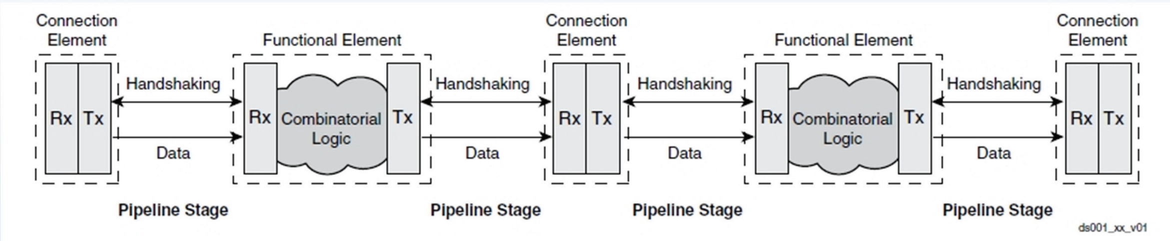
Achronix Semiconductor (http://www.achronix.com/)
–Ultra-fast (asynchronous picoPIPE) FPGAs
–Reliability in high radiation environments and over wide temperature range

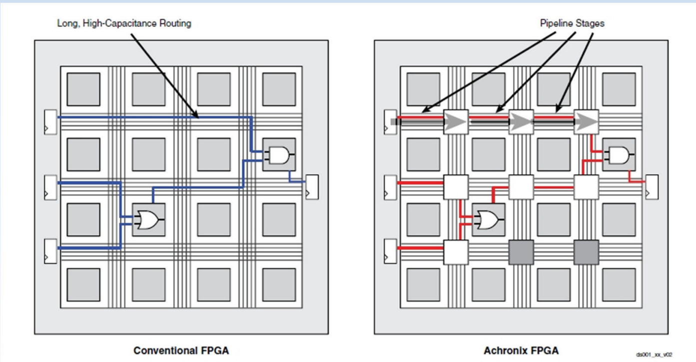
Tiempo Secure (www.tiempo-secure.com)
–IPs and EDA tools for the design of clockless ICs
–clockless microcontroller and clockless crypto-processor

Maxim Integrated (www.maximintegrated.com)
– asynchronous MCU for security applications

Many start-ups needed to shut down: Silistix Inc. (EDA tools for the design of customized on-chip interconnect using asynchronous circuits), Elastix Inc. (Generate asynchronous implementations of synchronous designs automatically), Handshake Solutions (spin-out of Philips, EDA tool and IP for asynchronous circuit design)
Few are successful:
Fulcrum Microsystems (Ethernet switch chips) acquired by Intel in 2011

picoPIPE Technology



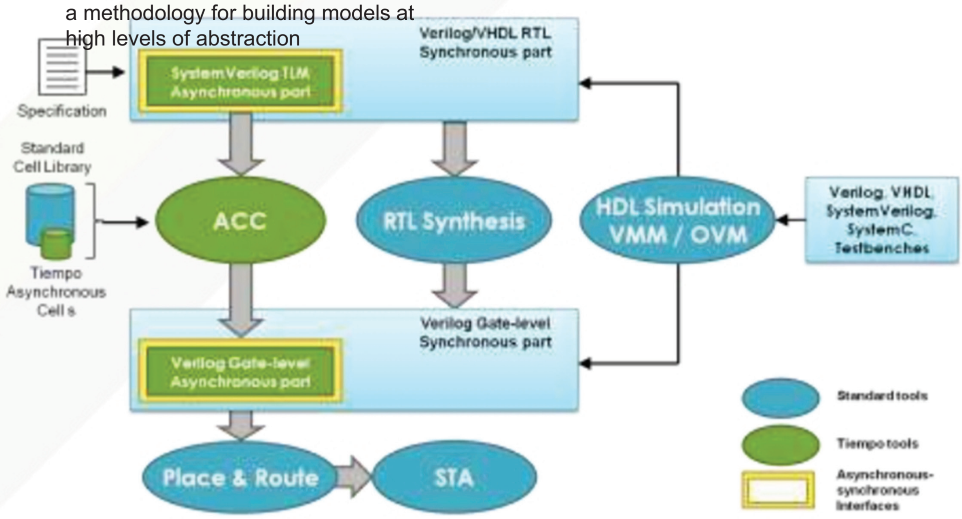


[Achronix Inc.]

The inherent pipelining of picoPIPE technology allows maximum throughput to be maintained, regardless of how large the FPGA is

Design Flow

Transaction-level modeling (TLM) is a methodology for building models at high levels of abstraction

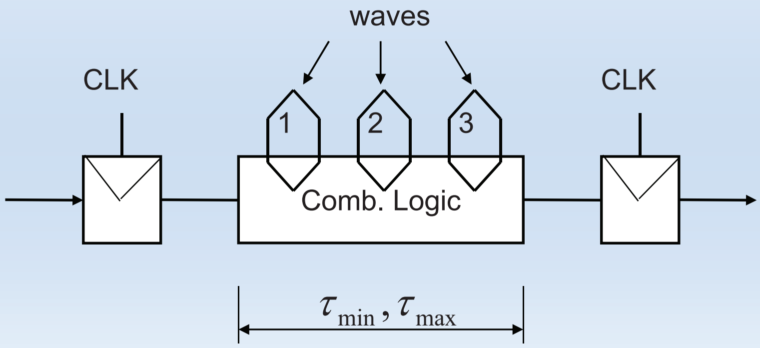


[Tiempo Inc.]

ACC (Asynchronous Circuit Compiler) is the first synthesis tool which automatically generates asynchronous circuits from a model written in a standard hardware description language.

Wave Pipelining

Pipelining is achieved by applying new data to inputs of comb. logic before computation of previous data is complete !



τ_{min}: minimum propagation delay of comb. logic

τ_{max}: maximum propagation delay of comb. logic

Wave Pipelining

No "crash" of data (wave) with previous data is allowed. So wave pipelined comb. logic only operate within a small frequency range!

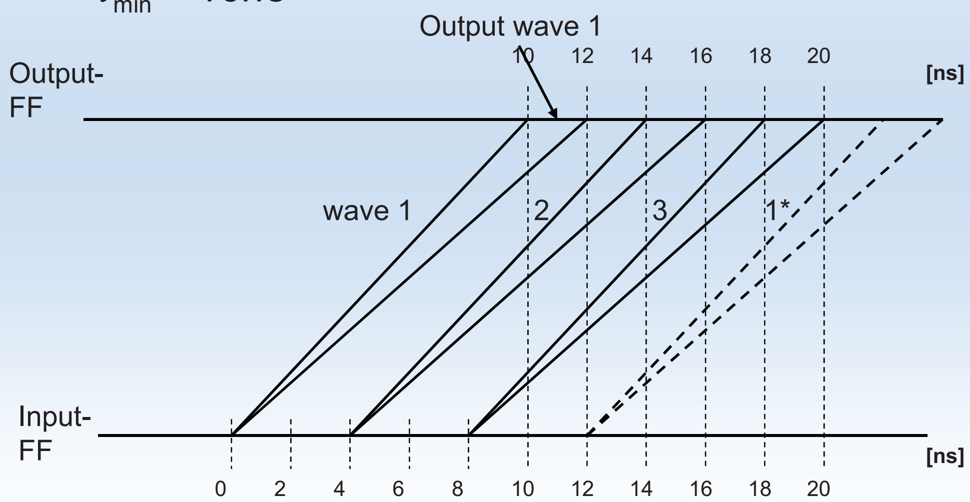
Constraint 1: after k-clock cycles data running on time critical path must be valid

Wave Pipelining

Ex: $K = 3$ number of waves (pipeline stages)

$\tau_{max} = 12ns$ **Constraint 1:** $\boxed{\tau_{max} \leq k \cdot T_{CLK}}$

$\tau_{min} = 10ns$



Wave Pipelining

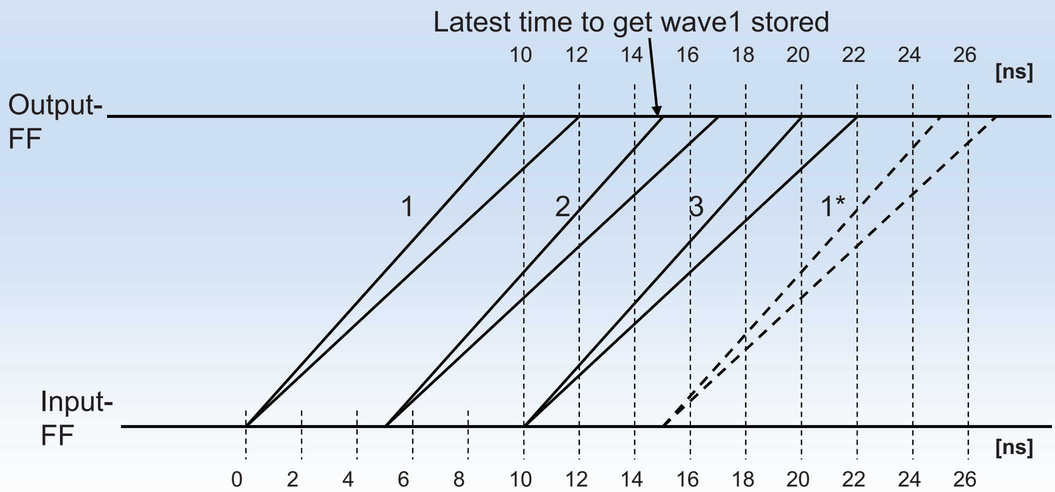
Constraint 2: after k-clock cycles data of the next wave running on the least time-critical path must not be valid at the output!

Wave Pipelining

Ex: $K = 3$ number of waves (pipeline stages)

$\tau_{\max} = 12ns$ **Constraint 2:** $\boxed{k \cdot T_{CLK} \leq T_{CLK} + \tau_{\min}}$

$\tau_{\min} = 10ns$



Wave Pipelining

Result: two-sided constraint on the clock period

$$\boxed{\frac{\tau_{max}}{k} \leq T_{CLK} \leq \frac{\tau_{min}}{K-1}}$$

- Valid clock period interval is maximum for $\tau_{min} = \tau_{max}$
  - Delay balancing required
  - Variations of supply voltage and temperature critical for wave pipelining